

Exam in “Economics & Law”

Winter Semester 2024/2025

Please read carefully the following instructions:

- This is a written exam based on 3 questions.
 - Each answer awards a different amount of points, as stated in brackets at the end of the question. The total points sum up to 100.
- This is an open-book exam. You may bring with you any written/printed material but you are not allowed to use a computer.

Questions:

Q1. Suppose there is a market for robots that is characterized by regular supply and demand curves. Three events occur one after the other: First, the government imposes a per-unit tax on the production of robots. Second, there is an increase in the demand for AI-processors, which are used as an input in the production of robots. Third, there is a decrease in the income of all consumers.

(a) Explain how this chain of events is expected to effect the equilibrium quantity, equilibrium price, and efficiency in the market for robots. **[15 points]**

(b) The government is considering imposing a lump-sum tax instead of a per-unit tax. If this occurred, how would this affect your answer to section (a)? **[5 points]**

(c) Suppose that Germany can produce either 1 Robot or 2 Cars in any given minute, but Canada can produce 3 Robots and 4 Cars at any given minute. Who should produce which product, and why? **[10 points]**

Q2. Adam and Ben are two friends who decide to set up a startup together. They need to decide who will be the CEO and who will be the COO and they agree on the following procedure: each person privately writes his name on a note and states “CEO” or “COO”. Then, they read the notes together. If one person write “CEO” whereas the other writes “COO”, then the CEO gets a yearly salary of 6 Million Euros, whereas the COO gets a salary of 2 Million Euros. If both write “COO” then the startup never gets formed, and they each get zero. If both write “CEO”, they become co-managers and each earn a salary of 4 Million Euros.

(a) Construct a payoff table. **[7 points]**

(b) State for each of the cells in the table whether it is Kaldor-Hicks Efficient and whether it is Pareto Efficient. **[8 points]**

(c) What is the Nash Equilibrium (or Nash Equilibria if there is more than one) of this game? **[5 points]**

(d) Suppose that the parties could bargain at zero transaction costs. How would this affect the outcome of the game? **[5 points]**

(e) Suppose that the negotiation would instead work as follows: Adam announces aloud first whether he wants to be CEO or COO. Then, Ben responds after hearing the declaration. The payoffs otherwise remain the same. How would this change affect the outcome of the game? **[5 points]**

(f) Suppose that Adam and Ben would only earn 1 Million as co-managers. How would this affect your analysis in the previous subsections? **[15 points]**

Q3. The “Republic of Nimbala” (a hypothetical country) is a resource-rich developing nation that emerged from a civil war three years ago. Despite its mineral wealth and agricultural potential, the country faces several challenges. The hastily negotiated peace agreement created a fragmented government where former war factions control different ministries. The Minister of Mining, a former warlord, requires all foreign companies to partner with local firms owned by his family members before receiving licenses. Only 30% of declared mining revenues reach the national treasure, despite some collection efforts. Furthermore, land ownership is contested between colonial-era title holders, indigenous communities claiming ancestral rights, and war refugees resettled by various government programs. Court backlogs for land disputes exceed five years. Transactions in movable assets (including cars, furniture, and others) are regulated by the housing commission, led by the president’s daughter, who is insisting that all transactions must be documented by carving the names of the parties into a wooden desk in her office. Moreover, business registration requires navigating 15 different government agencies, with each step demanding unofficial “speed processing fees.” Many officials maintain luxury properties abroad despite modest salaries. A recent microfinance program also revealed that poor people are unable to get access to credit, because the government’s “directed lending program” prohibits giving loans to anyone who cannot prove they have assets worth at least 5,000 Euros.

(a) Identify at least four distinct institutional and economic problems that Nimbala is facing. For each problem, explain the underlying causes and how it impedes economic development. **[16 points]**

(b) Propose a specific solution for three of the problems you identified in Part A. At least one solution must apply concepts from behavioral law and economics. For each solution, explain how it addresses the underlying causes and discuss potential implementation challenges **[9 points]**